Video 01 — Basic reinforcement learning: narration script and teaching outline

Video file: BlueImpulse_Video01_Basic_Reinforcement_Learning.mp4
Viewing position: 1 of 5
Duration: 10:00 (600 seconds)
Format: MP4, 1920 × 1080, 60 fps
File size: 47,425,849 bytes (47.43 MB)

Prerequisites: Basic arithmetic and the meanings of state and action. No programming experience is needed.

Learning goals:

- Name the learning loop
- Balance exploration and exploitation
- Calculate one Q-value update

Chapter and subtopic connections:

- Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop
- Chapter 2 — Behaviorism & Behaviorist AI / Rewards, policies & values
- Chapter 2 — Behaviorism & Behaviorist AI / Explore or exploit?
- Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value
- Chapter 2 — Behaviorism & Behaviorist AI / Training is different from using a policy

Apply in the learning system: Complete ACT-02 in the Study Guide / Markdown activity file. Use Test Bank questions TF05–TF08, MC05–MC08 and OE02 for review. The six-chapter PPT deck introduces the same chapter concepts through classroom discussion.

Optional interactive companion: ROVERLAB lets you apply these ideas by building a robot, connecting its network and comparing stimulus-response rules. Play at https://roverlab-blue-impulse.vercel.app and use the ROVERLAB activity in the Study Guide. It is a simplified rule-and-feedback model, not a neural network or the numerical Q-learning algorithm taught in Video 01.

Materials: Paper and a calculator are sufficient. The optional supporting Python demonstration uses Python 3.9 or newer with no third-party packages or account.

```
python3 supporting/BlueImpulse_Demonstrations.py --topic rl
```

Chapter timestamps and teaching route

Time	Teaching step	Chapter / subtopic
00:00–00:32	Pip's route is blocked	Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop
00:32–01:11	Today's route	Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop
01:11–02:15	Follow the learning loop	Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop
02:15–03:19	Reward, policy, value	Chapter 2 — Behaviorism & Behaviorist AI / Rewards, policies & values
03:19–04:11	The restaurant analogy	Chapter 2 — Behaviorism & Behaviorist AI / Explore or exploit?
04:11–04:53	Is the familiar route certainly best?	Chapter 2 — Behaviorism & Behaviorist AI / Explore or exploit?
04:53–05:38	What is a Q value?	Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value
05:38–06:36	Work through the worked update	Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value
06:36–07:45	A better estimate, not a proof	Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value
07:45–08:29	Try a terminal state	Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value
08:29–09:30	Experience changes estimates	Chapter 2 — Behaviorism & Behaviorist AI / Training is different from using a policy
09:30–10:00	Pip completes the shift	Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop

Using this script

The following narration is the exact measured cue text for the revised recording. Cue timestamps refer to the video timeline; small gaps include intentional thinking time and transition space. The voice is synthetic (Kokoro ONNX, am_michael), not a recording of a student. The full teaching outline states the purpose, visual focus and learner action alongside the words spoken.

00:00–00:32 — Pip's route is blocked

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop

Teaching / visual direction: Keep Pip’s problem visible while the opening question establishes why the concept matters. Key visual labels: PIP / ROUTE BLOCKED / TRY AGAIN.

00:00.150–00:02.347: Pip, the coffee is getting cold!

00:02.347–00:08.533: Our simulated campus rover has a delivery to make, but its familiar route is blocked.

00:08.533–00:17.133: Repeating yesterday's move will not solve today's problem. Pip needs useful feedback, and a way to change its estimates.

00:17.133–00:21.591: Have you ever wondered how an agent learns which action to try?

00:21.591–00:30.660: In Chapter two, Behaviorism and Behaviorist AI, we will connect the reinforcement-learning loop to one numerical update.

00:32–01:11 — Today’s route

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop

Teaching / visual direction: Reveal the learning goals in the order spoken, then retain a small progress cue. Key visual labels: LOOP / CHOICES / Q UPDATE.

00:32.498–00:39.921: Today you will identify an agent, an environment, a state, an action, a reward, and a policy.

00:39.921–00:47.984: You will compare exploration with exploitation, then calculate the example in the subtopic Update a Q value.

00:47.984–00:53.338: Welcome to the tutorial on basic reinforcement learning by Blue Impulse.

00:53.338–01:01.338: Have paper ready for one calculation. The tutorial PDF includes the same worked steps and practice question.

01:01.338–01:09.468: Treat our rover as an illustration. We are explaining a learning method, not claiming every real rover learns this way.

01:11–02:15 — Follow the learning loop

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop

Teaching / visual direction: Follow the spoken explanation with concise labels and the corresponding diagram state; transform the existing objects into the next scene. Key visual labels: STATE / ACTION / REWARD / NEXT STATE.

01:11.628–01:20.271: Start with the subtopic The reinforcement-learning loop. One interaction connects an observation, an action, and feedback.

01:20.271–01:27.097: The agent selects an action. The environment changes and returns the next state and a reward.

01:27.097–01:34.968: The learning algorithm uses that experience to improve the policy, which is the strategy for choosing actions.

01:34.968–01:40.536: For our grid, the state could be the current square, and an action could be moving right.

01:40.536–01:47.938: The reward is a number associated with the result. It is feedback, not a complete instruction manual.

01:47.938–01:54.102: Notice the return arrow. Learning depends on connecting an action to what happened after it.

01:54.102–02:02.617: For Pip, define the state before choosing an action. A location alone may be insufficient if a barrier can change the route.

02:02.617–02:12.220: The environment includes what happens after the action. The reward is feedback about that outcome, not a replacement for observing the next state.

02:15–03:19 — Reward, policy, value

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / Rewards, policies & values

Teaching / visual direction: Follow the spoken explanation with concise labels and the corresponding diagram state; transform the existing objects into the next scene. Key visual labels: REWARD / POLICY / VALUE.

02:15.280–02:21.914: The subtopic Rewards, policies and values separates three ideas that are easy to confuse.

02:21.914–02:28.442: Think of a journey. The reward scores an outcome now, while the policy chooses the next move.

02:28.442–02:34.798: A value estimate predicts how much return a choice may lead to, including later outcomes.

02:34.798–02:41.326: A route with a small cost now can still be worthwhile if it reaches a better destination.

02:41.326–02:47.981: That is why choosing the biggest immediate reward is not always the best long-term decision.

02:47.981–02:54.316: A poorly chosen reward can encourage a shortcut that misses the designer's real goal.

02:54.316–03:04.708: Suppose the reward gives a point for moving, regardless of direction. Pip could keep moving in circles and collect points without completing a delivery.

03:04.708–03:16.341: An objective and a reward design are connected, but they are not automatically identical. Test whether the rewarded behaviour is the behaviour you actually wanted.

03:19–04:11 — The restaurant analogy

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / Explore or exploit?

Teaching / visual direction: Follow the spoken explanation with concise labels and the corresponding diagram state; transform the existing objects into the next scene. Key visual labels: EXPLORE / EXPLOIT / COMPARE.

03:19.454–03:24.040: Use a restaurant to understand the subtopic Explore or exploit?

03:24.040–03:31.805: Ordering the dish you currently believe is best is exploitation. Trying a different dish is exploration.

03:31.805–03:37.479: Exploration gathers information because your current estimates might be wrong.

03:37.479–03:45.756: In our epsilon-greedy example, epsilon is zero point one, giving a ten percent exploration probability.

03:45.756–03:50.768: Otherwise, the agent chooses an action with the highest current Q value.

03:50.768–03:59.902: These percentages describe probabilities over choices, not a promise that exactly one out of every ten choices explores.

03:59.902–04:09.056: In a real physical system, exploration also needs safety constraints. A simulation is a useful place to experiment.

04:11–04:53 — Is the familiar route certainly best?

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / Explore or exploit?

Teaching / visual direction: Keep the supplied question legible during the thinking interval; reveal the answer only after the pause. Key visual labels: FAMILIAR / UNTRIED / EVIDENCE?.

Learner question: Two successes versus an untried route: proof?

Thinking interval: 04:23.232–04:28.232. The viewer may pause for longer.

04:11.628–04:17.452: Pause and think. One route has worked twice. Another route has never been tried.

04:17.452–04:23.232: Does that prove the first route is better? Take a few seconds before we reveal the answer.

04:28.232–04:34.354: No. The evidence is incomplete, so the familiar route is not certainly the better one.

04:34.354–04:42.396: Exploration may improve the estimate, but it has a cost. The design must balance information and performance.

04:42.396–04:50.950: Write one sentence explaining why exploitation uses estimated future return, rather than only the next reward.

Expected learning-check result: No. The evidence is incomplete.

04:53–05:38 — What is a Q value?

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value

Teaching / visual direction: Follow the spoken explanation with concise labels and the corresponding diagram state; transform the existing objects into the next scene. Key visual labels: OLD ESTIMATE / TARGET / UPDATE.

04:53.367–05:01.708: Now use the subtopic Update a Q value. Q associates a state and an action with an estimated return.

05:01.708–05:09.302: Q-learning is called model-free because it does not require a supplied model of environment transitions.

05:09.302–05:16.725: Model-free does not mean memory-free. The estimates still live in a table or another representation.

05:16.725–05:22.357: The update combines an old estimate with a target derived from new experience.

05:22.357–05:30.335: Alpha controls how far the estimate moves toward that target. Gamma discounts the future-value contribution.

05:30.335–05:36.179: We will keep every input visible, so you can reproduce the arithmetic yourself.

05:38–06:36 — Work through the worked update

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value

Teaching / visual direction: Follow the spoken explanation with concise labels and the corresponding diagram state; transform the existing objects into the next scene. Key visual labels: OLD Q = 2 / TARGET = 4.6 / NEW Q = 3.3.

05:38.759–05:45.051: Start with an old Q value of two and a learning rate, alpha, of zero point five.

05:45.051–05:51.131: The reward is one, gamma is zero point nine, and the best next Q value is four.

05:51.131–05:58.511: First calculate the target: one plus zero point nine times four. The target is four point six.

05:58.511–06:05.615: Next subtract the old estimate. Four point six minus two gives an error of two point six.

06:05.615–06:11.310: Multiply that error by alpha. Half of two point six is one point three.

06:11.310–06:17.795: Finally add the change to the old value. Two plus one point three gives three point three.

06:17.795–06:25.819: Watch the estimate move halfway toward the target. That halfway movement is what alpha equals one half means here.

06:25.819–06:33.669: The supporting Python file repeats this calculation and prints every intermediate value for you to inspect.

06:36–07:45 — A better estimate, not a proof

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value

Teaching / visual direction: Follow the spoken explanation with concise labels and the corresponding diagram state; transform the existing objects into the next scene. Key visual labels: ONE EXPERIENCE / BETTER ESTIMATE / NO GUARANTEE.

06:36.672–06:44.436: This experience raised the estimate from two to three point three. Now decide what that result actually means.

06:44.436–06:51.668: It did not prove that the action is always best. One observation is not a reliable policy.

06:51.668–06:58.408: Repeated experience, an appropriate learning schedule, and enough exploration are still important.

06:58.408–07:05.021: There is also a special case: when the next state is terminal, the future-value term is zero.

07:05.021–07:11.186: That prevents the calculation from inventing future reward after the episode has ended.

07:11.186–07:19.078: Compare these two ideas: updating an estimate from evidence, and guaranteeing the outcome of the next action.

07:19.078–07:24.048: They are different claims, and only the first follows from our calculation.

07:24.048–07:33.139: Imagine that the next trial gives different feedback. One sample is noisy evidence, so another update can move the estimate again.

07:33.139–07:42.592: The learning rate controls how strongly this experience changes the estimate. It does not turn a single successful delivery into a guarantee.

07:45–08:29 — Try a terminal state

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / Update a Q value

Teaching / visual direction: Keep the supplied question legible during the thinking interval; reveal the answer only after the pause. Key visual labels: TERMINAL STATE / NO FUTURE RETURN / NEW Q = 1.5.

Learner question: Terminal state: old Q = 2, alpha = 0.5, reward = 1

Thinking interval: 07:58.414–08:04.414. The viewer may pause for longer.

07:46.063–07:51.908: Your turn. Keep the old value at two, alpha at one half, and the reward at one.

07:51.908–07:58.414: Make the next state terminal. What is the new Q value? Pause the video if you need more time.

08:04.414–08:12.114: The future contribution is zero, so the target is one. The error is one minus two, or negative one.

08:12.114–08:19.218: Half that error is negative one half. Add it to two, giving a new value of one point five.

08:19.218–08:27.711: The estimate falls because the observed target is lower. A learning update can decrease a value as well as increase it.

Expected learning-check result: New Q = 2 + 0.5 × (1 − 2) = 1.5

08:29–09:30 — Experience changes estimates

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / Training is different from using a policy

Teaching / visual direction: Follow the spoken explanation with concise labels and the corresponding diagram state; transform the existing objects into the next scene. Key visual labels: OBSERVE / ACT / LEARN / TRY.

08:29.889–08:37.312: Return to Pip's blocked route. The agent acts, receives feedback, and learns a policy from experience.

08:37.312–08:47.790: Reward scores an outcome. Value estimates return. Exploration improves information, while exploitation uses the best current estimates.

08:47.790–08:55.021: Our worked Q update produced three point three, and our terminal-state practice produced one point five.

08:55.021–09:03.216: The training-and-deployment distinction matters: running an already learned policy is not the same as training it online.

09:03.216–09:08.335: A fixed sensor-response rule can control a machine without learning at all.

09:08.335–09:16.462: Before moving on, explain the loop to someone else and run the included calculation with one changed reward.

09:16.462–09:27.686: Pip's improvement is a change in an estimate and, possibly, a later choice. It is not proof that every new street or unexpected obstacle has been solved.

09:30–10:00 — Pip completes the shift

Content reference: Chapter 2 — Behaviorism & Behaviorist AI / The reinforcement-learning loop

Teaching / visual direction: Resolve the opening problem and use a concise Blue Impulse end title, returning to the opening composition. Key visual labels: BLUE IMPULSE / FOLLOW THE SIGNAL.

09:30.933–09:34.857: Pip's small problem now has an explanation you can test.

09:34.857–09:42.750: Use Chapter 2, Behaviorism & Behaviorist AI, and the named subtopics to revisit the reasoning.

09:42.750–09:52.203: The tutorial guide includes the full teaching steps and the checked practice answers. Run the supporting demonstration and change one input.

09:52.203–09:57.707: Blue Impulse. Follow the signal, predict the result, and check your explanation.

Supporting demonstration and limits

Use old Q = 2, alpha = 0.5, reward = 1, gamma = 0.9 and best next Q = 4. Alpha is the learning rate and gamma discounts future return. The step-by-step arithmetic is:

Step	Calculation	Result
Target	1 + 0.9 × 4	4.6
Error	4.6 − 2	2.6
Change	0.5 × 2.6	1.3
Updated value	2 + 1.3	3.3

Practice: Keep old Q, alpha and reward unchanged but make the next state terminal. Predict the new value before reading on.

Checked answer: The future contribution is zero. Target = 1; error = −1; change = −0.5; new Q = 1.5. An estimate may fall after evidence as well as rise. Neither update proves that the next action will succeed.